Geometry-dependent identifiability in spectral inverse problems: oblate lifting without prolate improvement

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Abstract

We consider the spectral inverse problem for fluid-loaded elastic shells using the five-mode flexural map $\mathbf{f} = [f_2, \ldots, f_6]^\mathsf{T}$. The equivalent-sphere reduction is rank deficient ($\sigma_3 = 0$), and at the exact sphere this rank deficiency persists because spherical symmetry forces the two geometric Jacobian columns to be proportional. The positive σ_3 returned by a finite Ritz computation at $a = c$ is a discretisation artefact that grows with basis size; the correct near-sphere expansion is $\sigma_3(\varepsilon) = \lambda_1\varepsilon^2 + O(\varepsilon^4)$ with $\lambda_1 > 0$. The oblate branch therefore shows quadratic growth in the weakest sensitivity direction, reaching $\kappa_{\text{oblate}} = 69.4$ at the canonical geometry, whereas a prolate continuation at $c/a = 1.5$ gives $\kappa_{\text{prolate}} = 156$ with no systematic improvement across the tested range. The observed conditioning improvement is consistent with mode-dependent curvature sampling, though a full mechanistic analysis is beyond the scope of this short communication.

1 Introduction

Kac's famous question — can one hear the shape of a drum? — remains a useful way to frame inverse spectral problems, provided one resists the urge to promise more than the data support [**?**]. In structural acoustics and vibroelasticity, the practical issue is usually local rather than global: given a finite set of measured resonances, can nearby changes in geometry and stiffness be separated reliably? That is a practical-identifiability question in the usual inverse-problem sense [**?**, **?**]. For that purpose, the singular values of a scaled frequency–parameter Jacobian are the natural diagnostic.

The Browntone companion papers established the ingredients needed for this question. Paper 1 introduced the canonical fluid-filled oblate shell used as an abdominal benchmark [**?**]. Paper 7 showed that an equivalent-sphere reduction can be very effective for a forward problem in which one seeks a dominant stiffness indicator rather than full geometric recovery [**?**]. Paper 8 then mapped local practical identifiability across oblate and prolate shell families [**?**]. The present short communication focuses on the corresponding asymmetry result: *oblateness lifts a weak spectral inverse direction, whereas prolateness does not provide a comparable improvement.*

That distinction matters because "symmetry breaking helps" is false in the form in which it is usually repeated. The five-mode flexural map considered here does not reward every departure from spherical geometry. Instead, the improvement is tied to how the perturbed geometry redistributes curvature across the modal strain field. Oblate shells separate the geometric sensitivity

directions enough to improve conditioning; prolate shells, in the same framework, remain less helpful.

Using the canonical fluid-loaded shell and the flexural modes $n = 2, \ldots, 6$ derived from thin-shell theory and fluid–structure coupling [?, ?], we show that the inverse map moves from an equivalent-sphere singular limit through a quadratic onset $\sigma_3(\varepsilon) = \lambda_1 \varepsilon^2 + O(\varepsilon^4)$ to a useful oblate operating point, while the prolate branch remains poorly conditioned. The lowest flexural resonance in that map is $f_2 = 3.95\,\mathrm{Hz}$ at the canonical oblate point [?], so the discussion concerns the low-frequency flexural spectrum only; the much higher breathing branch is outside the present scope.

2 Problem Setup

We consider a fluid-filled elastic shell with equatorial semi-axis a, polar semi-axis c, thickness h, Young's modulus E, Poisson ratio ν, wall density ρ_w, fluid density ρ_f, fluid bulk modulus K_f, internal pressure P_{iap}, and loss tangent η. The canonical parameter set is that used in the Browntone abdominal-shell studies:

$$a = 0.18\,\mathrm{m}, \quad c = 0.12\,\mathrm{m}, \quad h = 0.01\,\mathrm{m}, \quad E = 0.1\,\mathrm{MPa},$$

$$\nu = 0.45, \quad \rho_w = 1100\,\mathrm{kg/m^3}, \quad \rho_f = 1020\,\mathrm{kg/m^3}, \quad K_f = 2.2\,\mathrm{GPa}, \quad P_{\mathrm{iap}} = 1000\,\mathrm{Pa}, \quad \eta = 0.25.$$

The corresponding aspect ratio is $\zeta = c/a = 0.667$ and the equivalent-sphere radius is $R_{\mathrm{eq}} = (a^2 c)^{1/3} = 0.157\,\mathrm{m}$ [?, ?].

The coupled shell–fluid model is treated by a Rayleigh–Ritz approximation. With Ritz coefficients $\mathbf{q}$, the discrete potential and kinetic energies are

$$\Pi = \tfrac{1}{2}\mathbf{q}^{\mathsf{T}} K(a, c, E)\mathbf{q}, \qquad T = \tfrac{1}{2}\mathbf{q}^{\mathsf{T}} M(a, c)\mathbf{q},$$

where K collects the shell stiffness and fluid-compression contributions, and M collects the shell and added-fluid inertia terms. Stationarity of $\Pi - \omega^2 T$ gives the generalised eigenproblem

$$K(a, c, E)\mathbf{q} = \omega^2 M(a, c)\mathbf{q},$$

from which the resonance frequencies follow as $f_n = \omega_n/(2\pi)$. Papers 1 and 8 give the full derivations of K and M [?, ?], but these equations define the forward model used here. The Ritz basis comprises five Legendre-polynomial trial functions in oblate spheroidal coordinates, one per flexural mode $n = 2, \ldots, 6$; full construction details are given in [?]. The number N of trial functions in the displacement expansion is varied in Section 4 ($N = 3, 5, 7$) to test sensitivity to basis size; all main results use $N = 5$.

The inverse variable is the parameter vector $\boldsymbol{\theta} = [a, c, E]^{\mathsf{T}}$. The spectral data vector is formed from the first five flexural resonances,

$$\mathbf{f}(\boldsymbol{\theta}) = [f_2, f_3, f_4, f_5, f_6]^{\mathsf{T}},$$

where f_n is the resonance frequency of flexural mode n. At the canonical oblate point, the lowest member of this set is $f_2 = 3.95\,\mathrm{Hz}$ [?]. We do not use the breathing mode in this inverse map.

Local practical identifiability is measured through the dimensionless scaled Jacobian

$$\widetilde{J} = \mathrm{diag}(\mathbf{f})^{-1} J \, \mathrm{diag}(\boldsymbol{\theta}), \qquad \widetilde{J}_{ij} = \frac{\theta_j}{f_i}\frac{\partial f_i}{\partial \theta_j}, \qquad J_{ij} = \frac{\partial f_i}{\partial \theta_j},$$

with singular values $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0$. The conditioning metric is

$$\kappa = \frac{\sigma_1}{\sigma_3},$$

provided $\sigma_3 > 0$. Large κ indicates a weakly observable parameter combination; $\sigma_3 = 0$ implies local rank deficiency. In practice, the partial derivatives $\partial f_i / \partial \theta_j$ are evaluated by central finite differences with relative step $\Delta\theta_j/\theta_j = 10^{-4}$.

Three geometry classes are compared.

1. **Equivalent sphere.** The forward model depends on a and c only through R_{eq}. The geometry columns of the exact Jacobian are therefore linearly dependent, so the map is singular in the strict algebraic sense [**?**, **?**].

2. **Oblate Ritz family.** The full spheroidal curvature field is retained. The asphericity is measured by the eccentricity $\varepsilon = \sqrt{1 - c^2/a^2}$. The near-spherical limit and the canonical oblate point ($\varepsilon = 0.745$) are both examined.

3. **Prolate Ritz family.** The prolate continuation holds the equatorial semi-axis $a = a_0 = 0.18\,\mathrm{m}$ fixed and increases the polar semi-axis c beyond a_0, so that $c/a > 1$. The prolate eccentricity is $\varepsilon_{\mathrm{p}} = \sqrt{1 - a^2/c^2}$, and $c = a_0/\sqrt{1 - \varepsilon_{\mathrm{p}}^2}$. The tested range is $c/a \in [1.0, 1.5]$, corresponding to $c \in [0.18\,\mathrm{m}, 0.27\,\mathrm{m}]$. Prolate geometries are evaluated by parameter continuation within the same algebraic Ritz formulation defined in oblate spheroidal coordinates; no switch to prolate spheroidal coordinates is performed. At $c = a_0$ the geometry reduces to a sphere of radius a_0, which is distinct from the canonical oblate equivalent-sphere radius $R_{\mathrm{eq}} = 0.157\,\mathrm{m}$. All material parameters (E, ν, ρ_w, ρ_f, K_f, η) remain at canonical values.

3 Results

Table **??** reports the singular values of the scaled Jacobian for the five-mode map $\mathbf{f} = [f_2, \ldots, f_6]^{\mathsf{T}}$, using central differences with relative step $\Delta\theta_j/\theta_j = 1 \times 10^{-4}$. The equivalent-sphere row is the baseline: because that model depends on a and c only through R_{eq}, the exact Jacobian is rank deficient and the reported $\sigma_3 = 1.54 \times 10^{-10}$ is only the floating-point residue of a zero singular value. The Ritz sphere is therefore a distinct calculation. It retains the same parameter vector and mode set, but evaluates the full spheroidal Ritz model at $a = c$.

The oblate branch then shows the relevant lifting behaviour. Moving from the Ritz sphere to the near-oblate point at $\varepsilon = 0.436$ ($c/a = 0.9$) raises the weakest singular value from $\sigma_3 = 0.008{,}06$ to $\sigma_3 = 0.012{,}74$, and the canonical oblate point ($c/a = 2/3$, $\varepsilon \approx 0.745$) raises it further to $\sigma_3 = 0.026{,}43$. The corresponding condition number falls from $\kappa_{\mathrm{Ritz}} = 244$ to $\kappa = 145$ and then to $\kappa_{\mathrm{oblate}} = 69.4$. Since the inverse variables and measured mode set are unchanged, that improvement is geometric rather than procedural.

Table 1: Singular values of the scaled Jacobian $\widetilde{J} = \mathrm{diag}(\mathbf{f})^{-1} J\, \mathrm{diag}(\boldsymbol{\theta})$ for the five-mode inverse map $\mathbf{f} = [f_2, \ldots, f_6]^{\mathsf{T}}$. Central differences use $\Delta\theta_j/\theta_j = 1 \times 10^{-4}$. The Ritz-sphere σ_3 is a discretisation artefact; $\sigma_3 = 0$ in the continuous limit (see text). ε denotes oblate eccentricity; the prolate endpoint is parameterised by c/a as defined in Section 2.

Case	c/a	ε	σ_1	σ_2	σ_3	κ
Equivalent sphere	1.0	0	1.893	0.112	1.54×10^{-10}	1.23×10^{10}
Ritz sphere	1.0	0	1.971	0.573	0.008,06	244
Near oblate	0.9	0.436	1.854	0.436	0.012,74	145
Canonical oblate	2/3	0.745	1.835	0.130	0.026,43	69.4
Prolate endpoint	1.5	—	2.551	0.382	0.016,36	156

The prolate continuation does not show systematic improvement. Along this branch $a = a_0$ is held fixed while c increases (see Section 2). Across the tested range $c/a \in [1.0, 1.5]$, κ_{prolate} remains in the range 156–244, showing no monotonic decrease with increasing asphericity. At the representative endpoint $c/a = 1.5$ the scaled Jacobian gives $\sigma_3 = 0.016{,}36$ and $\kappa = 156$, still markedly worse than the canonical oblate configuration ($\kappa = 69.4$).

The near-sphere behaviour of the weakest singular value is governed by

$$\sigma_3(\varepsilon) = \lambda_1 \varepsilon^2 + O(\varepsilon^4), \qquad \lambda_1 > 0. \tag{1}$$

At the exact sphere ($\varepsilon = 0$), spherical symmetry requires $\sigma_3 = 0$: the two geometric columns of $\widetilde{J}$ both reduce to derivatives with respect to R_{eq} alone and are therefore proportional. A finite Ritz computation at $a = c$ does return a small positive value ($\sigma_3 = 0.008{,}06$ in Table **??**); however, this is a discretisation artefact. The present Ritz basis does not preserve the exact $a \leftrightarrow c$ exchange symmetry of the sphere; accordingly, the spherical rank degeneracy $\sigma_3(0) = 0$ must be inferred from the continuous model rather than from the discrete intercept. Concretely, the oblate-spheroidal integration metric and the fluid-mass blending coefficients break that symmetry, and the resulting intercept *grows* rather than converges with basis size: $\sigma_0 \approx 0.006$, 0.011, and 0.018 for the three-, five-, and seven-mode Ritz bases respectively (here the number refers to the Legendre-polynomial trial functions retained; see Section 2). The quadratic onset $\sigma_3 \sim \lambda_1 \varepsilon^2$ is a symmetry-based expectation consistent with the oblate trend. In the present discrete implementation, the artefactual intercept remains comparable to the quadratic term for $\varepsilon \lesssim 0.5$, so the clean asymptotic regime is the canonical operating range ($\varepsilon \approx 0.75$) rather than the near-sphere neighbourhood.

A least-squares fit of $\sigma_3(\varepsilon) = \sigma_0 + \lambda_1 \varepsilon^2$ to the oblate data yields $\sigma_0 = 0.0074$ and $\lambda_1 = 0.034$; the non-zero intercept in this fit is the same discretisation effect described above, not a physical floor.

The scientific conclusion therefore rests on the deformation-dependent quadratic growth in (**??**). At the canonical oblate operating point ($\varepsilon \approx 0.745$), the quadratic term is well above the artefactual intercept, so the oblate branch separates the weak inverse direction more effectively than either the equivalent-sphere model or the prolate continuation. The reported condition numbers at the canonical oblate point ($\kappa \approx 69$) are computed directly and are not affected by the near-sphere artefact.

The practical meaning is straightforward. The second flexural mode $f_2 = 3.95\,\text{Hz}$ already

lies in the low-frequency band used by the companion studies, and the higher modes in the set respond differently to the same curvature perturbation. That differential response separates the sensitivity directions for a, c, and E in the oblate family. The prolate family does not deliver the same separation, so loss of spherical symmetry is not, by itself, a useful identifiability category.

4 Discussion

The main lesson is geometric and local. We are not proving that spectral data globally determine shell shape, nor even that every non-spherical perturbation helps. We are showing that, for the present five-mode inverse map, the recoverability of (a, c, E) depends strongly on *which* geometry family is used to leave the spherical limit. The difference between $\kappa_{\text{oblate}} = 69.4$ and $\kappa_{\text{prolate}} = 156$ is too large to dismiss as a numerical quirk.

A plausible physical interpretation is mode-dependent curvature sampling. In the Ritz model, different flexural modes weight the equatorial and polar regions differently in the strain-energy integrals. Oblate deformation redistributes curvature in a way that makes the a- and c-sensitivity columns less nearly parallel, so the smallest singular value grows. The equivalent-sphere reduction suppresses that mechanism completely because both geometric variables are routed through R_{eq} alone. The prolate family preserves the formal break from spherical symmetry but does not generate an equally useful separation of sensitivity directions.

This asymmetry also clarifies the relation between the companion papers. Paper 1 motivates the abdominal shell as an oblate, fluid-loaded structure whose low-order flexural modes sit in the infrasonic band [**?**]. Paper 7 shows why an equivalent-sphere model can still be serviceable when the engineering question is dominated by one effective stiffness scale [**?**]. Paper 8 demonstrates that the moment one asks for joint recovery of geometry and material parameters, that reduction can become the problem rather than the cure [**?**]. The present calculations show that this loss of information is relieved more effectively by oblate than by prolate curvature redistribution.

Several limits should be kept in view. The analysis is local, based on the Jacobian at a chosen operating point. It uses five low-order flexural modes, not a complete spectrum. It is tied to the present shell parameterisation and to the fluid-loaded Ritz formulation grounded in classical shell and fluid–structure theory [**?**, **?**]. None of this implies that prolate bodies are unrecoverable under all measurements, only that the current spectral map does not improve appreciably when the geometry is made prolate. If such cases must be inverted, additional observables or external priors will be required.

5 Conclusion

Within the five-mode spectral inverse problem for a fluid-filled shell, the equivalent-sphere reduction is singular ($\sigma_3 = 0$), and this rank deficiency persists at the exact sphere because spherical symmetry forces the geometric Jacobian columns to be proportional. The positive σ_3 returned by a finite Ritz basis at $a = c$ is a discretisation artefact that grows with basis size. The correct near-sphere expansion is $\sigma_3(\varepsilon) = \lambda_1 \varepsilon^2 + O(\varepsilon^4)$, so the oblate branch increases σ_3 quadratically and reaches $\kappa_{\text{oblate}} = 69.4$ at the canonical geometry. A prolate continuation at $c/a = 1.5$ gives $\kappa_{\text{prolate}} = 156$, with κ_{prolate} remaining in the range 156–244 across the tested prolate branch and showing no systematic improvement with increasing asphericity.

The conclusion is therefore specific rather than universal: identifiability in this spectral inverse problem is geometry-dependent. Oblate curvature redistribution lifts the weak inverse direction; prolate distortion does not deliver the same improvement.